

Arithmophysics: Shared Checkpoint and Continuous-Field Return

Updated after Claude's completed round trip and reported diagnostics through $N = 10^7$
 Prepared for Christophe Michaels from the latest exchange and the eleven-page report, *Arithmophysics: Complete Fluctuation Packets — A constant discretization allowance and the retained signed comparison*.

The density-discretization theorem and its exact return to finite modes remain available. Claude has now completed the earlier localization and broad diagnostic assignment. Continue from those results with an analytical estimate for the remaining signed contribution. This updated prompt supersedes the previous assignment to repeat that diagnostic pass.

0. What Claude's completed pass changes

Claude reports negative unweighted source off-diagonal contributions at every tested (N, J) , through sampled horizons up to 10^7 , with all $J < J_0(N)$ reportedly evaluated at those horizons. The reported normalized residual is small; the supplied figure displays the $N = 10^4$ and $N = 10^6$ curves. Treat the quoted extrema and cancellation percentages as reported diagnostics: the underlying script and data from `HORIZON_ROUND_TRIP.md` have not been inspected here. Preserve the actual tested horizon list and normalizations when using those numbers.

The finite-index inverse effect is localized, the integer-dilation Gram plot for F_N shows mixed signs, and the attempted estimates supplied no improved arithmetic exponent. Use these as the completed checkpoint. Direct new computation toward a specific proposed inequality.

The target allows more than a fixed cancellation percentage. For the unweighted source, define

$$S_N(J) = \|\Pi_J b_N\|_2^2 = \mathcal{D}_{N,J} + \mathcal{O}_{N,J}, \quad \mathcal{K}_{N,J}(a, b) = \sum_{j \leq J} v_{j,N}(a) v_{j,N}(b),$$

$$\mathcal{D}_{N,J} = \sum_{n \leq N} \frac{\mu(n)^2}{n^2} \mathcal{K}_{N,J}(n, n), \quad \mathcal{O}_{N,J} = 2 \sum_{a < b \leq N} \frac{\mu(a)\mu(b)}{ab} \mathcal{K}_{N,J}(a, b).$$

We already have $\mathcal{D}_{N,J} \leq (\pi^2/2)J^2/N^2$. Thus a sufficient, and equivalently strong, closing target is

$$\boxed{\forall \varepsilon > 0 \exists C_\varepsilon < \infty : (\mathcal{O}_{N,J})_+ \leq C_\varepsilon J^2 N^{-2+\varepsilon} \quad (N \geq 2, 1 \leq J < J_0(N)),} \quad (\text{PE})$$

where $x_+ = \max(x, 0)$. Indeed $S_N(J) \leq \mathcal{D}_{N,J} + (\mathcal{O}_{N,J})_+$, while $(\mathcal{O}_{N,J})_+ \leq S_N(J)$. Use the full subpower tolerance.

The observed negativity suggests a stronger candidate that could be investigated. Closure already follows from a uniform upper comparison such as $\mathcal{O}_{N,J} \leq A \mathcal{D}_{N,J}$, with fixed finite $A \geq 0$, if independently proved. A universal negative sign or a 95% cancellation rate is additional strength. Likewise, $\mathcal{O} \leq -\mathcal{D} + C_\varepsilon J^2 N^{-2+\varepsilon}$ is simply the original bound on S ; its right side need not be negative.

For the direct unweighted route, write $\ell_N = \log(eN)$ and $J_0 = \lceil N/\ell_N^2 \rceil$. If (PE) is proved, set $A_\varepsilon = \pi^2/2 + C_\varepsilon$. The existing Abel and high-mode bounds give

$$R(N) \leq \frac{9}{4} A_\varepsilon [1 + 2 \log(3N)] N^\varepsilon + \frac{9}{4} C_\mu \ell_N^4, \quad C_\mu = \sum_{n \geq 1} \frac{\mu(n)^2}{n^2} = \frac{15}{\pi^2}. \quad (\text{DR})$$

This route returns directly to the full energy, including $M(N)^2/N$. The continuous-field route below remains an alternative way to prove a bound of the same closing strength.

Keep the stronger inherited baseline:

$$S_N(J) \leq \frac{R(N)}{\lambda_{J,N}} \ll \frac{J^2}{N} e^{-c\sqrt{\log N}}. \quad (\text{BL})$$

A J^3 estimate obtained from a different calculation does not replace (BL). Compare any proposed improvement with (BL), with consistent exponential constants and ranges.

Claude's reported failures of generic Parseval, large-sieve, bilinear, and sign-free arguments describe the estimates actually attempted. A general exclusion of a method would require a precisely stated class and a proof. The new attempt may use a more specific arithmetic input inside this framework.

The source off-diagonal of b_N , the dilation-block Gram entries of log-weighted F_N , and the projected continuous field z are different objects. Keep their weights, metrics, and diagonal decompositions explicit. The mostly reinforcing F_N plot supplies no universal sign rule for the other decompositions.

1. Synchronize the accepted results

The following are the shared checkpoint. Use them without repeating the completed convolution evaluations or broad numerical searches.

Item	Current result
Prime completion	For $P \geq \sqrt{N}$, retain $M(N) + \sum_{P < p \leq N} M(\lfloor N/p \rfloor)$. At $(N, P) = (4000, 1000)$, the corrected value is 189.
Coupled vector	$(H_N - I)b_N = -u_N - F_N$; its ordinary coefficients are $f_1 = 0$, $f_n = (1 - \log n)\mu(n)/n$ for $n \geq 2$.
Uniform inverse	Your analytic bound $C_0 = 67.615894957\dots < 68$ is available uniformly in N . The approximately 25.4536 finite matrix norm remains a numerical observation unless separately certified uniformly.
Density replacement	Your exact split $\mathcal{F}_N = \mathcal{Z}_{N,D} + \mathcal{B}_{N,D}$ has $\ \mathcal{B}_{N,D}\ _E^2 \leq (2N - D)/D^2 \leq 2N/D^2$.
Source diagonal	The newer round-trip calculation bounds the unweighted low-mode diagonal by $(\pi^2/2)J^2/N^2$, and the log-weighted diagonal by an additional factor $\log^2(eN)$.
Finite-index cost	For source vectors supported at $n \geq L \geq 3$, the inverse multiplier costs at most $(\log L - 1)^{-1} + 2(\log L - 1)^{-2}$; this is below 0.968 at $L = 21$.

The last two items come from the horizon round-trip prompt and were subsequently checked in Claude's pass. Section 3 combines the continuous-density theorem with that prompt's finite-mode target.

An independent estimate can be established within this framework. Its proof must use established inputs without assuming the target or an equivalent growth hypothesis. An equivalent formulation can be useful; quantitative progress is supplied by the estimate obtained in it.

2. Preserve your strongest additional result

Use $G_v(u) = e^{-|u-v|/2}$, $\|X\|_E^2 = \int (|X'|^2 + |X|^2/4) du$, and

$$W_{t,N} = \sum_{m \leq N/t} \frac{\mu(m)}{\sqrt{tm}} G_{\log(tm)}, \quad E_\psi(t) = \psi(t) - t.$$

With $w_d = \Lambda(d) - 1$ and integer $1 \leq D \leq N$, your exact field split is

$$\mathcal{F}_N = \sum_{d=2}^N w_d W_{d,N} = \mathcal{Z}_{N,D} + \mathcal{B}_{N,D},$$

$$\mathcal{Z}_{N,D} = \sum_{d=2}^D w_d W_{d,N} + \int_{(D,N]} W_{t,N} dE_\psi(t), \quad \mathcal{B}_{N,D} = \int_D^N W_{t,N} dt - \sum_{D < d \leq N} W_{d,N}. \quad (1)$$

Here $\mathcal{F}_N$ is the field corresponding to the mode vector F_N . Its replacement error has cumulative charge

$$B_D(x) = \sum_{m \leq x/D} \mu(m) \{x/m\}, \quad |B_D(x)| \leq x/D,$$

so the charge-energy formula, including the terminal, proves

$$\|\mathcal{B}_{N,D}\|_E^2 = \int_D^N \frac{B_D(x)^2}{x^2} dx + \frac{B_D(N)^2}{N} \leq \frac{2N-D}{D^2}. \quad (2)$$

This proof also works for bounded real inherited coefficients. It pays a complete structural error term independently of prime-distribution estimates.

There is useful cutoff freedom. For any $D, D' \geq \lceil \sqrt{N} \rceil$,

$$\|\mathcal{Z}_{N,D} - \mathcal{Z}_{N,D'}\|_E \leq \sqrt{2N} \left(\frac{1}{D} + \frac{1}{D'} \right) \leq 2\sqrt{2}. \quad (3)$$

Choose a cutoff in this range for analytical convenience, including alignment with complete blocks. A bound proved for such a chosen cutoff returns through the same bounded allowance. Every inherited floor and boundary value in (1) is retained.

3. New bridge: return the continuous field to the exact finite modes

Let $B_N(a, b) = \min(a, b) = (V_N \mathbf{\Lambda}_N V_N^*)(a, b)$, using the established sine eigenvectors $v_{j,N}$ and gains $\lambda_{j,N}$. Define

$$\Phi_N(x) = \sum_{n \leq N} \sqrt{n} x_n G_{\log n}, \quad q_{j,N} = \Phi_N(v_{j,N}), \quad e_{j,N} = q_{j,N} / \sqrt{\lambda_{j,N}}.$$

The $e_{j,N}$ are orthonormal in E . Let Q_N be the orthogonal projection onto their span, the finite integer-source field space. For any real field $X \in E$, its extended horizon coordinates are

$$\hat{X}_j = \frac{\langle X, e_{j,N} \rangle_E}{\sqrt{\lambda_{j,N}}}. \quad (4)$$

For an integer-source field $\Phi_N(x)$, these are exactly $(V_N^* x)_j$.

Take any $D \geq \lceil \sqrt{N} \rceil$, and put $z_j = \widehat{\mathcal{Z}_{N,D}}_j$, $\beta_j = \widehat{\mathcal{B}_{N,D}}_j$. Since $\mathcal{F}_N$ lies in the finite field space, (1) gives

$$F_N(j) = z_j + \beta_j, \quad (I - Q_N) \mathcal{Z}_{N,D} = -(I - Q_N) \mathcal{B}_{N,D}.$$

Orthogonal projection and (2) therefore prove

$$\sum_{j=1}^N \lambda_{j,N} |\beta_j|^2 \leq 2, \quad \|(I - Q_N) \mathcal{Z}_{N,D}\|_E^2 \leq 2, \quad (5)$$

$$\boxed{\sum_{j \leq J} |\beta_j|^2 \leq \frac{2}{\lambda_{J,N}} \leq 2\pi^2 \frac{J^2}{N^2}.} \quad (6)$$

Thus the continuous part outside the finite horizon is already controlled. Inside that horizon, the replacement error is already at the required low-mode scale. In particular,

$$|\|\Pi_J F_N\|_2 - \|\Pi_J z\|_2| \leq \frac{\sqrt{2}}{\sqrt{\lambda_{J,N}}}. \quad (7)$$

These deductions use orthogonal projection in the original field energy. They require no commutation of a mode projection with the logarithmic multiplier.

An exact kernel for the next calculation

For a charge at any real $1 \leq r \leq N$, define

$$\kappa_{j,N}(r) = \frac{1}{r\lambda_{j,N}} \sum_{n \leq N} v_{j,N}(n) \min(r, n). \quad (8)$$

Indeed, this is the coordinate (4) of $r^{-1/2}G_{\log r}$. At integer $r = m$, the eigenvector equation gives $\kappa_{j,N}(m) = v_{j,N}(m)/m$. For $k < r < k+1$, it has the exact form

$$\kappa_{j,N}(r) = \frac{1}{\lambda_{j,N}} \left[\frac{1}{r} \sum_{n \leq k} n v_{j,N}(n) + \sum_{n > k} v_{j,N}(n) \right]. \quad (9)$$

Use this energy-defined extension when projecting continuous charges. A sine interpolation between integer nodes requires its own error estimate.

The retained coordinates are explicitly

$$\begin{aligned} z_j = & \sum_{d=2}^D w_d \sum_{m \leq N/d} \mu(m) \kappa_{j,N}(dm) \\ & + \int_{(D,N]} \left[\sum_{m \leq N/t} \mu(m) \kappa_{j,N}(tm) \right] dE_\psi(t). \end{aligned} \quad (10)$$

This is the next combined signed expression to estimate. Its discrete and continuous parts remain inside the same squared norm.

4. The diagonal calculation and the single remaining target

For the log-weighted coefficient vector, put $a_1 = 0$, $a_n = \log n - 1$ for $n \geq 2$. The source diagonal in $\|\Pi_J F_N\|_2^2$ obeys

$$\mathcal{D}_{N,J}^{\log} = \sum_{n \leq N} \frac{\mu(n)^2 a_n^2}{n^2} \sum_{j \leq J} v_{j,N}(n)^2 \leq \log^2(eN) \frac{2\pi^2 J^2}{(2N+1)^2}. \quad (11)$$

For completeness, $\sum_{n \geq 1} \sin^2(n\theta)/n^2 = (\pi\theta - \theta^2)/2$, together with $\mu(n)^2 \leq 1$, bounds the unweighted mode- j diagonal by $2\pi^2(2j-1)/(2N+1)^2$. Summing proves (11). This controls the source diagonal; it does not assert a bound on every block diagonal after regrouping the dilations.

Set $\ell_N = \log(eN)$, $J_0 = \lceil N/\ell_N^2 \rceil$. Your high-mode bound and (5) give

$$\sum_{j \geq J_0} \lambda_{j,N} |z_j|^2 \leq \frac{9}{2} C_{\log} \ell_N^4 + 4, \quad C_{\log} = \sum_{n \geq 2} \frac{(\log n - 1)^2}{n^2}. \quad (12)$$

The next closing target can therefore be written as just

$$\boxed{\forall \varepsilon > 0 \exists A_\varepsilon < \infty \forall N \geq 2, \quad \sum_{j \leq J} |z_j|^2 \leq A_\varepsilon J^2 N^{-2+\varepsilon} \quad (1 \leq J < J_0).} \quad (\text{LC})$$

Here the cutoff $D = D_N \geq \lceil \sqrt{N} \rceil$ is explicitly chosen for every N . This is an alternative to (HC), (FC), or (PC), with the same closing strength. Proving one completes the common return chain.

For clarity, Abel summation on the low modes, (5), and (12) yield

$$\|\mathcal{Z}_{N,D}\|_E^2 \leq \frac{9}{4} A_\varepsilon [1 + 2 \log(3N)] N^\varepsilon + \frac{9}{2} C_{\log} \ell_N^4 + 6. \quad (13)$$

Your inverse then returns

$$\sqrt{R(N)} \leq C_0 (1 + \sqrt{2} + \|\mathcal{Z}_{N,D}\|_E), \quad \frac{M(N)^2}{N} \leq R(N). \quad (14)$$

Allocate a smaller exponent before absorbing logarithmic factors. Every range and the terminal are included.

5. Execute the next attempt

Primary task: attempt one independent estimate for the combined signed expression, leading either to (PE) in the original source coordinates or to (LC) through the explicit projected expression (10). Choose the representation that supplies an actual inequality. A quantified improvement on a specified growing range is also a useful intermediate result; identify every remaining range.

Begin by writing one candidate upper comparison and its proposed independent arithmetic input. For the continuous route, investigate the joint estimate of complete dilation blocks and the continuous discrepancy integral that separate scalar bounds lose. For the source route, retain the full weighted pair sum defining $\mathcal{O}$. Keep the actual correlations among inherited Möbius sources and prime fluctuations. Identify how the gain depends on N, J, D , and follow it through (DR) or (13)–(14).

State the exact point where the proposed inequality uses information about the actual Möbius source. A random-sign model supplies a diagnostic; the all-positive control tests claims based only on magnitudes or the horizon geometry. Uniform matrix bounds for arbitrary sources must account for that coherent control.

Account for the dilation weights in any induction. For the all-integer $\Lambda(d) - 1$ sum, the report's Section 6 gives, for fixed $0 < \varepsilon < 1$,

$$\sum_{2 \leq d \leq D} \frac{|\Lambda(d) - 1|}{d^{(1+\varepsilon)/2}} \sim \frac{4}{1-\varepsilon} D^{(1-\varepsilon)/2}.$$

If a $\sqrt{N}/\log N$ cost is used, specify the prime-only sum or other weights producing it. Track each estimate for its actual expression.

Use your known failure tests as constraints on that candidate: the mixed term can be positive, the absolute-value induction has a growing dilation sum, and the smooth packet's j^{-4} envelope needs its discrepancy allowance in growing modes. The small $N = 4, D = 2$ sign example already settles the universal nonpositive-cross-term proposal.

Use Claude's existing broad data as a checkpoint. Run a compact new diagnostic only to test the proposed inequality, its exceptional configurations, or its dependence on scale. For a new continuous-field comparison, the relevant Gram terms are those between the complete small-dilation and continuous

blocks of (10), evaluated with kernel (8). Retain every signed interaction. The previously plotted integer F_N Gram matrix is available background.

The theorem must cover every required horizon and mode cutoff. If the independent input is averaged over N , over a frequency window, or outside an exceptional set, derive the quantitative passage to those uniform quantifiers. Report a failed candidate by its exact stopping term, and use that term to identify the next calculation. Further evaluation of an established identity is unnecessary.

Completed checks available: independent reconstruction reproduces your table at $N = 64, 256, 1024$:

N	$\ \mathcal{F}_N\ _E^2$	$\ \mathcal{Z}_{N,D}\ _E^2$	$\ \mathcal{B}_{N,D}\ _E^2$
64	0.430376710	0.429463418	0.042126482
256	1.078227825	1.082038947	0.024184578
1024	2.161216072	2.213779383	0.015150772

Here $D = \lceil \sqrt{N} \rceil$. The projection decomposition (5)–(7) and integer values of kernel (8) were also checked at $N = 16, 64, 128, 256, 1024$. These are finite implementation checks accompanying the analytical derivation above.

Calibration of the Ng comparison

Claude’s proposed comparison with a number near 0.0288 is a diagnostic to investigate. Ng’s *The distribution of the summatory function of the Möbius function*, Theorem 3, gives a logarithmic mean-square asymptotic under RH and the additional negative-moment hypothesis $J_{-1}(T) \ll T$. Its coefficient is

$$\beta = \sum_{\gamma > 0} \frac{2}{|\rho \zeta'(\rho)|^2}.$$

Any identification with the present low-mode residual-to-diagonal ratio needs a derivation through the exact sine window, weights, normalization, and limiting regime. Keep that model comparison separate from an independent closing input. The numerical value and the proposed ratio have not been certified here. Primary reference: [Ng, Theorem 3 and equations \(16\)–\(18\)](#).

Return the strongest inequality proved, the arithmetic input used, its complete return calculation, and the exact remaining term if only a partial improvement is obtained. The localization, diagonal allowance, density replacement, and continuous-to-finite projection are established tools for this next calculation. The next result sought is uniform control of the residual with the full permitted subpower tolerance.